A Survey on Secure and Private Federated Learning Using Blockchain: Theory and Application in Resource-Constrained Computing

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***Abstract*—Federated learning (FL) has gained widespread popularity in recent years due to the fast booming of advanced machine learning and artificial intelligence, along with emerging security and privacy threats. FL enables efficient model generation from local data storage of the edge devices without revealing the sensitive data to any entities. While this paradigm partly mitigates the privacy issues of users' sensitive data, the performance of the FL process can be threatened and reach a bottleneck due to the growing cyber threats and privacy violation techniques. To expedite the proliferation of the FL process, the integration of blockchain for FL environments has drawn increasing attention from academia and industry. Blockchain has the potential to prevent security and privacy threats with its decentralization, immutability, consensus, and transparency characteristics. However, if the blockchain mechanism requires costly computational resources, then the resource-constrained FL clients cannot be involved in the training. Considering that, this survey focuses on reviewing the challenges, solutions, and future directions for the successful deployment of blockchain in resource-constrained FL environments. We comprehensively review variant blockchain mechanisms suitable for the FL process and discuss their trade-offs for a limited resource budget. Furthermore, we extensively analyze the cyber threats that could be observed in a resource-constrained FL environment, and how blockchain can play a key role in blocking those cyberattacks. To this end, we highlight some potential solutions for the coupling of blockchain and FL that can offer high levels of reliability, data privacy, and distributed computing performance.**

***Index Terms*—Blockchain, federated learning (FL), privacy, resource limitations, security.**

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I. Introduction

A. Motivation, Comparison, and Contributions

DATA-DRIVEN technologies can be limited by factors, such as limited computing resources or the need for large amounts of high-quality data. Earlier online methodologies transferred raw data which created potential data privacy risks. Data silos form in organizations rendering them inaccessible to other departments and often incompatible with data sets within the same organization. Federated learning (FL) is introduced as a promising solution for privacy-preserving and scalable machine learning [1]. In FL, machine learning model parameters are exchanged instead of raw data that preserves user data privacy to some extend [1], [2]. Centralized computing architectures can be directly targeted and denied due to their single-point-of-failure vulnerability. In contrast, the decentralized nature of blockchain spreads control across a network of nodes, greatly lowering the likelihood of a security breach. The immutability of data stored on a blockchain also ensures that information added cannot be changed or removed without the network's consent. In particular, blockchain being a decentralized approach can improve system resilience against adversaries. FL and blockchain technology together offer a synergistic approach that combines the benefits of both paradigms to produce a powerful solution with special advantages. The combination of FL and blockchain has increased robustness compared to earlier methodologies that exhibited diminishing performance, due to features like a lack of privacy preservation. While blockchain's tamper-proof and transparent nature assures that model updates and transactions are stored immutably, FL enables training models on decentralized data sources without disclosing raw data. Besides, due to blockchain's transparency, any participant may confirm the accuracy of model changes and the consensus process, minimizing the possibility of bias or malicious manipulation. This is especially useful in fields, such as healthcare and finance, where transparency and accountability are paramount. The combination of FL and blockchain technology shows great promise in environments, where processing resources, network connectivity, and data availability may be constrained.

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This survey examines blockchain-based FL as a solution for privacy preservation and secure resource management. Recent blockchain and FL surveys did not comprehensively analyze the combination of the two to provide a secure learning setting for a resource-constrained environment.

In this article, we present the following contributions.

1) We conduct a comprehensive analysis of the security elements of Blockchain and FL in resource-constrained computing contexts. We focus on investigating the possibilities of blockchain technology to protect FL's security and privacy.
2) We categorize various cyber-attack models and point out the leading vulnerabilities that could be observed during various stages of the FL process and in online environments. Additionally, we identify promises of blockchain-enabled FL to deal with these challenges.
3) We present the existing blockchain-based FL applications and highlight potential solutions to the existing research challenges, with a focus on resource-constrained computing environments.
4) We introduce promising future research for securing technologies at the intersection of blockchain and FL. The key insight is that proposed decentralized opportunities offer a new perspective on the resource constraints of various participating devices.

B. Organization of the Survey

The remainder of this article is organized as follows. In Section II, we present an overview and taxonomy of FL and blockchain with a comprehensive list of existing studies. In Section III, we review distributed ML approaches. Section IV examines applications of blockchain FL. Section V presents a detailed analysis of the major challenges of FL when applied to resource-constrained devices, and potential solutions to those emerging challenges. Finally, in Section VI we conclude our article.

II. Background Study

Online computing environments can put private information at risk when communicating online. FL can help ensure data privacy throughout communications [3]. FL transmits user data through machine learning updates instead of raw data. Classical data-sharing techniques can be improved by FL and its privacy protection mechanisms. Blockchain, a decentralized approach, offers additional security and resources for applications such as FL. Blockchained FL (BCFL) is an appropriate solution for resource-constrained computing environments due to peer-to-peer computing, network participation incentives, validation protocols, and other resource-preserving mechanisms [4]. The following section details mechanisms found within FL and blockchain.

A. Brief Introduction to Federated Learning

FL was introduced by McMahan et al. [1] in 2016, proposing it as a method based on data parallelism. FL safeguards user privacy by enabling devices to collectively train an ML model through collaborative use of local data and model parameters. FL can process model updates synchronously or asynchronously. FL is a collaborative machine-learning architecture that stores local machine learning parameters and their data on a device. This local data reduces the need for raw data to be transmitted or stored online in a cloud. Wei et al. [5], stated that FL, as a manner of distributed machine learning, can significantly preserve clients' private data from being exposed to external adversaries [5]. FL environments have an objective for user data to be unidentifiable while preserving good qualities for machine learning model optimization. differential privacy (DP), a privacy-preserving technique is an approach that adds random noise to individual's data in such a way that individual's data is indistinguishable from the group of data if captured. DP increases the privacy guarantee on the user end and is applicable to any data set.

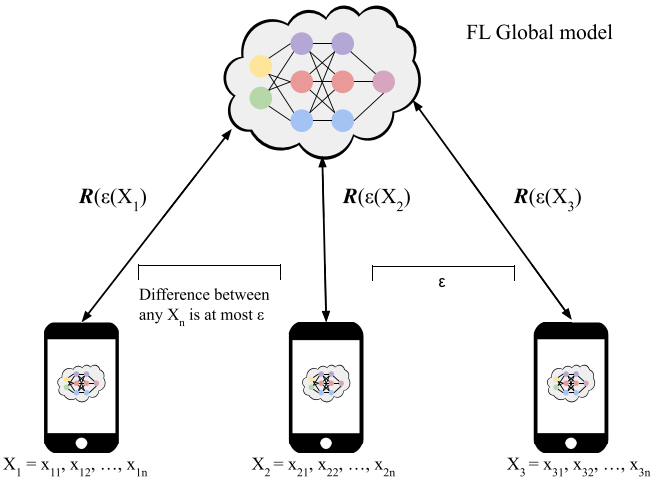


Fig. 1. DP within an FL environment. DP applies ϵ to each participant to guarantee that the maximum distance between any participant X_n is the same.

Independent and identically distributed random variable data (IID) seen in Fig. 1 can safeguard FL environments from inference attacks. IID data is considered homogeneous, meaning data samples share the same data properties. In contrast, heterogeneous data distributions are increasingly difficult to capture data findings due to including diverse data, imbalanced data, and even noisy incomplete data.

Data distributions encourage distinct FL implementations for optimal data management. Each FL structure looks at feature spaces and data sets differently. personalized FL (PFL) is a proposed solution for the challenge of approaching heterogeneous data structures through multimodel methods. Firdaus et al. [6] implemented a blockchain PFL architecture with personalized local training models and personalized global model aggregation. FL implementations can benefit from combining edge computing and blockchain mechanisms. Three different types of FL implementations include Horizontal FL [7], Vertical FL [8], [9], and Federated Transfer Learning [10], [11], [12], [13], [14]. Traditional FL has a point of failure limitation. Fig. 1 showcases an uninterrupted online learning environment, although outsiders deploying a denial of service could halt the repeated process. Internet of Things (IoT) devices differ in performance and reliability. The FL training process could be computationally expensive depending on available resources, which may require devices to drop

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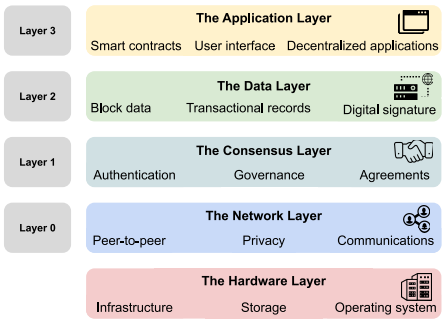


Fig. 2. Layers within a Blockchain architecture. An example deployed blockchain architecture can consist of the Solidity programming language to handle smart contracts and block transaction records, Hyperledger to handle the distributed ledger and consensus, VPN NPSabbrpl for private communications, and reliable hardware to ensure participation and blockchain interoperability.

out from the learning process. Devices may need motivation to participate in FL training, which can be encouraged by blockchain incentives.

B. Brief Introduction to Blockchain

Blockchain is a revolutionary distributed and decentralized digital ledger technology that enables several parties to maintain a shared database without the need for centralized authority. It allows for secure and transparent data capture, storage, and verification. The blockchain technology was first established as the underlying technology for the cryptocurrency Bitcoin, but it has subsequently found applications far beyond the realm of finance, such as supply chain management, healthcare, digital identity verification, voting systems, and many more. It operates across a network of computers, referred to as nodes, with each node holding a copy of the blockchain. The blockchain comprises of a series of blocks, each filled with relevant transaction history, such as unique identifiers, block computing costs, and other pertinent network details. Network transactions are directly recorded onto the blockchain once they are authenticated while peer-to-peer transactions allow workloads to share resources and information. Peers in the network can operate as both resource suppliers and consumers. Imteaj et al. [15] used the phrases 'participants' and 'miners' to denote the nodes or agents in a blockchain network. Participants are the agents in charge of carrying out transactions, whereas miners are in charge of block validation. Concurrent network processing occurs in the context of a blockchain, resulting in a large optimization surface area. Li et al. [16] and Hieu et al. [17] highlighted some crucial factors, e.g., processor power, storage space, and available network bandwidth that need to be considered while managing resources in resource-constrained computing settings.

When numerous activities run concurrently within a blockchain, performance might suffer, especially when blocks are generated quickly. This can lead to merge problems, which arise when two or more blocks with the same hash information are successfully mined simultaneously. Merging issues can create forks in the blockchain, causing alternative chains to emerge until solved by consensus. Having a proper consensus protocol can discourage multiple versions of a blockchain, and protecting the integrity of the blockchain allows historical data to sync efficiently. A rule of thumb is to reference the largest chain of blocks as the main blockchain to deter conflicts [18].

Fig. 2 displays a deployed blockchain architecture example. Blockchain architectures contain several layers as follows.

1) The hardware layer is necessary for hosting a blockchain architecture. The core infrastructure requires computers, graphics processing units (for miners), and miscellaneous data storage. Computing resources can be emulated through virtual machines as a process.
2) The network layer is responsible for peer-to-peer communication. Block transaction data is publicly visible in a public ledger. As a result, public blockchain architectures can share processes with unknown participants that can directly interact with one another.
3) The consensus layer authenticates transactions based on agreed-upon protocols. This layer is responsible for validating transactions. If a fork in the blockchain emerges, the consensus layer handles disputes using protocol logic.
4) The data layer includes blocks and block transaction records. Blocks specifically contain information about: the previous block hash, the timestamp of the block creation, the generated block hash, and the transaction nonce, which is used for verifying transactions.
5) The application layer contains the front end of the blockchain and is considered the visible layer for the user. This layer contains an application programming interface (API), displays smart contracts, and includes basic configuration for quickly exchanging information with other blockchains. Blockchain interoperability allows similar blockchain structures to communicate efficiently.

1) Blockchain Fundamentals: Blockchain architectures use cryptographic hash functions for transactions. Miners solve cryptographic puzzles to generate blocks and may form mining pools to boost performance. Nonce (numbers used only once) are numbers that miners must calculate before solving a cryptographic block. Cryptographic hash functions are encrypted, requiring efficient computing techniques to reasonably solve them. The consensus layer authenticates that block conditions are met before integrating blocks into the blockchain. Blockchain agreements are transacted through smart contracts, rule-based digital agreements between two parties (i.e., miners and participants) that automatically execute when conditions are met. Once a block is authenticated, a new block is sought for mining. Imteaj et al. [15], Zhang et al. [19], and Iansiti an Lakhani [20] discussed that typically in a blockchain environment, constant creation of new blocks, even with no transaction, is crucial for maintaining security as it prevents malevolent users from creating longer, tampered blockchains [15].

2) Blockchain Categories: Generally, there are three types of blockchain: 1) public; 2) private; and 3) consortium. Each type of blockchain involves different permissions and participation specifics. Different types of consensus

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TABLE I
SUMMARY OF INCLUDED PAPERS REGARDING FL, BLOCKCHAIN, AND BCFL

Ref No.	Year	Contribution	Resource-constrained considerate	Participation incentivized
[26]	2022	Adaptive FL survey for minimizing resource consumption within communication-constrained environments. AFAFed approach reduces FL training time by up to 50%, although participation incentives are not considered.	Yes	No
[27]	2021	Comprehensive FL survey on security and privacy. Authors examine FL limitations of both security and privacy at the same time and countermeasures. Introduces the incorporation of artificial incentives, such as reputation.	No	No
[28]	2022	Blockchain FL model for blade icing estimation. Discusses how Blockchain can incentivize the enthusiasm of participants. Mentions PoS Blockchain mechanism to avoid high resource consumption during competition [29].	Yes	Yes
[30]	2022	Proposes a fast Blockchain-based communication efficient FL framework by compressing communications Employs the Top_k algorithm to compress model local model updates on each client to reduce communication load.	Yes	No
[31]	2021	Blockchain-empowered decentralized horizontal FL framework that improves 5G-Enabled UAV privacy. Authors consider the authentication problem for resource-constrained IoT devices.	Yes	No
[15]	2021	Comprehensive book on fundamental topics related to Blockchain. The book discusses energy prosumers which can also contribute in generating energy in addition to consuming. Prosumers who have extra energy resources can sell to other consumers to receive some incentives. States IoT devices are resource-constrained in nature.	Yes	Yes
[4]	2021	Comprehensive FL survey that considers IoT nodes as clients with limited resources challenges. Authors examine FL IoT resource constraints, i.e., selecting clients based on only resource availability can lead to choosing a malicious client [4].	Yes	Yes
[32]	2022	Comprehensive Blockchained FL survey for securing IoT devices. Authors mention blockchain and distributed ledger technology requiring considerable storage resources that are impossible to directly store on current IoT devices [32].	Yes	Yes
[33]	2021	Comprehensive public blockchain survey for resource-constrained IoT devices. Existing public blockchains are examined as well as challenges with integrating existing public blockchains with resource-constrained IoT devices.	Yes	Yes
[34]	2019	Authors propose a Blockchain FL (BlockFL) architecture and examine the end-to-end latency of BlockFL.	Yes	Yes
[35]	2022	Comprehensive Blockchained FL survey, where authors examine current feasible applications for BCFL in industrial applications. Several Blockchain consensus mechanisms and their merit and weaknesses were compared and detailed.	Yes	Yes
[36]	2021	Blockchained FL paper which explores integration opportunities with FLchain and mobile edge computing. Authors examine 4 key domains in FLchain design. Possible directions for further improving resource allocation domain include: jointly considering block generation [37], model arrival rate, and on-device learning rate [36].	Yes	Yes
[38]	2022	FL survey on applications within smart healthcare. The survey includes advanced FL designs for smart healthcare, including resource-aware FL, secure-FL, privacy-enhanced FL, incentive-aware FL, and personalized FL [38].	Yes	Yes
[39]	2019	FL paper that proposes end-to-end private FL system which uses secure multiparty computation (SMC) in combination with DP to produce models with high accuracy. The paper examines FL privacy guarantees and various trust scenarios.	No	No
[40]	2019	FL paper that proposes FL theoretical bounds, and a control algorithm to achieve the desirable trade-off between local update and global aggregation in order to minimize the loss function under a resource budget constraint [40].	Yes	No
[9]	2019	FL survey regarding concepts and applications. The authors introduce incentive mechanisms for commercial FL. Blockchain introduced as a potential solution for FL data management, incentives, and permanent data recording mechanism.	Yes	Yes
[41]	2020	Privacy-preserving FL paper that leverages smart home customers data through IoT devices to assist home appliance manufacturers. The approach includes a reputation-based crowdsourcing incentive mechanism for motivating reliable customers. The limited computing power of IoT devices and potential of edge computing considered.	Yes	Yes

and authority mechanisms are included in each blockchain type.

1) *Public Blockchain:* The most well-known blockchain is the public blockchain which is permissionless and open to all participants. Public blockchain ledgers are publicly available. Each participant has equal permissions in this peer-to-peer networking environment. Open participation in the blockchain has been known to affect the number of resources in many ways. Cases, in which open participation leads to a large number of dishonest participants, can cause blockchain performance to diminish. Examples of popular public blockchain include cryptocurrency platforms, such as Etheruem or Bitcoin.
2) *Private Blockchain:* Private blockchains are commonly referred to as permissioned blockchains. Only authorized participants are allowed to join and view private blockchain ledgers. The central entity of the private blockchain has the power to maintain protocols, validate a block, and alter any blockchain protocol (e.g., block consensus) [21], [22].
3) *Consortium Blockchain:* Consortium blockchains include public and private blockchain qualities, which are partially decentralized, permissioned and transparent [23], [24]. In consortium blockchain, selected authorities have increased permissions compared to the public. Increased permissions for consortium authorities include the ability to participate in block validation. Qi et al. [25] utilized a consortium blockchain for preselecting a limited number of trusted miners to maintain the distributed ledger of an intelligent transportation system. Consortium or permissioned blockchains are suitable when the integrity of participants is in question. Dishonest participants may sabotage others to compete for incentives or resources.

C. Comparing Existing Works

Table I summarizes various papers that consider participation incentives and resource constraints. When comparing existing surveys we examined prior works for being resource-constrained considerate and participation incentivized. Blockchain was an important factor for optimal resource allocations, due to providing decentralized resource opportunities to participants. Surveys that examined compressed communications and other distributed optimization approaches were evaluated as resource-constrained considerations. Prior works discussed ideal participants for optimal FL performance, however incentives for encouraging optimal participation were often undefined. Surveys that investigated different rewards for participants being honest, responsive, and original were highlighted. Both metrics from existing works are ongoing research due to trivial considerations such as fairness. IoT devices as clients are prone to online attacks, therefore reviewing how online attacks influence devices in terms of consensus, aggregation, and network efficiency are

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of interest. The following section examines potential attacks, vulnerabilities, and threats found in BCFL environments.

D. Threat Models and Integration Motivation

FL includes privacy protection mechanisms against unauthorized access to data. FL environments are prone to insider and outsider attacks from adversaries. Insider attacks occur within the FL environment and can take the form of a dishonest participant, honest-but-curious participant, or even an untrusted (malicious) FL server [42]. Outsider attacks are external attacks that target the FL network, such as a denial of service or hacker eavesdropping on communications. Malicious outsiders who have power or influence over an insider may coerce an otherwise benign insider [43]. Meticulous attack attempts can breach FL security in different ways. Table II briefly summarizes FL attacks and resource costs. Attacks can occur during device training, parameter upload and download, central aggregation, and post-aggregation phases. A range of various attacks can exploit FL vulnerabilities during each stage. Fig. 3 illustrates a classification of different FL attacks and threats.

1) Inference Attacks: During inference attacks, an adversarial agent attempts to capture sensitive information throughout training. Training data, participant data, and label data are recorded without permission in hopes of correlating encrypted information with real values. Generative adversarial networks (GANs) can create powerful inference attacks. Inference attacks fall into five general categories.

1) *Membership Inference Attack:* Attackers determine if an individual was present in the training data set. Shokri et al. [44] constructed a membership inference attack that exploits the observation that machine learning models often behave differently on the data they were trained on versus the data they see for the first time [44]. Adversaries attempt to infer from specific samples that can be captured from model outputs. The objective is for adversaries to determine if a specific record exists in the model training data set.
2) *Data Properties Inference Attack:* Data properties of a machine learning model are inferred using the parameters shared during model training. The particular data property is fully investigated to learn the frequency of the property within the training data. For example, Ganju et al. [45] demonstrated that a classifier that recognizes smiling faces also leaks information about the relative attractiveness of the individuals in its training set [45]. For example, a data properties inference attack on a model trained to screen FL participants by threshold may mistakenly reveal hardware or host types within the FL architecture.
3) *Data Samples and Labels Inference Attack:* An adversary uses inference to capture targeted data samples and labels. FL model classes and participant training input labels can be reconstructed for inference. A malicious participant calculates distributed gradients to infer private information about other participants or model parameters. Choquette-Choo et al. [46] demonstrated a label-only inference attack that could capture private information from machine learning models without access to confidence scores.
4) *Model Inversion Attack:* Model inversion attacks aim to reconstruct private information from training data. Attackers with access to model parameters can look at the model's confidence score of predicted classes for inference. Successful model inversion attacks infer realistic representations from training data. Fredrikson et al. [47] demonstrated the applicability of model inversion attacks on decision trees for lifestyle surveys as used on machine-learning-as-a-service systems and neural networks for facial recognition [47].
5) *Model Extraction Attacks:* Adversary attacks that aim to steal exact model parameters. Hu and Pang [48] mentioned how model extraction attacks aim to duplicate a machine learning model through query access to a target model. Hu and Pang studied two attacks on GANs: fidelity extraction attacks and accuracy extraction attacks [48]. A strategic position can be extracted from learned functionality. Transfer learning attacks can occur when attacks are trained on vulnerabilities of similar models with the same framework, and then transferred to attack the target model. Extracting knowledge of the target models' training data and functionality increases future attack effectiveness.

2) Poisoning Attacks: Poisoning attacks include misleading data injected by adversaries as inputs. Poisoning attacks can be classified into two categories: 1) model poisoning and 2) data poisoning attacks.

1) *Model Poisoning Attacks:* In model poisoning attacks, the attacker reduces the model's performance on targeted subtasks (e.g., classifying planes as birds) by uploading "poisoned" updates [49]. Poisoning attacks change the model's weights and biases, leading to the wrong classifications of data. The model's confidence decreases when interacting with poisoned data.
2) *Gradient Manipulation Attack:* Transmitted gradients and parameters are manipulated throughout the FL training process to reduce global model performance. As a result, global model accuracy is reduced by adversaries injecting malicious updates. In addition, wrong classifications can result from gradient manipulation, allowing adversaries to increase the success rate of their attacks.
3) *Training Rule Manipulation Attack:* An adversary attempts to minimize the difference between correct and incorrect training updates during training. The strength of this attack increases when multiple participants become dishonest and poison their parameters. Additionally, adversaries may bribe or collude with multiple participants to increase the targeted model's wrong classifications and lower detection mechanisms. The objective is for eventual harmful updates to go undetected based on the similarity of correct and incorrect training updates.
4) *Data Poisoning Attacks:* Training data is altered to impact the FL model negatively. During training,

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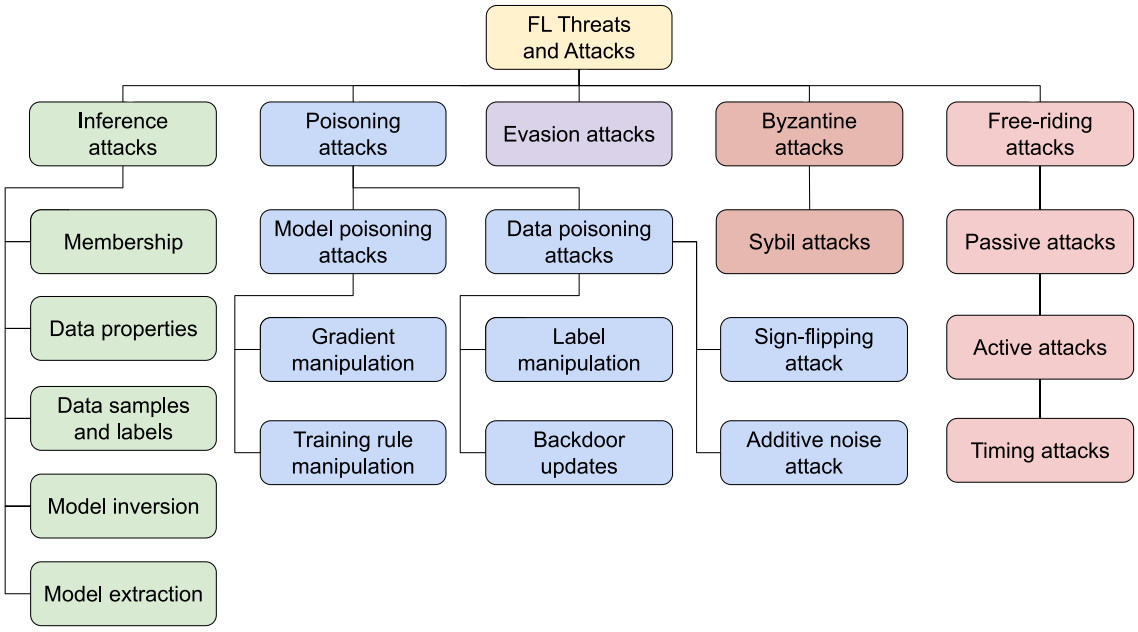


Fig. 3. Classifying FL threats and attacks.

contaminated data is introduced to corrupt the central aggregator. Data poisoning attacks inject malicious data into the training data set before the learning process starts [50], [51]. Such an attack allows adversaries to disguise harmful data, and eventually, the large amounts of poisoned data increase training times.

5) *Label Manipulation Attack:* Label manipulation attacks cause the model to mislabel training data. Label flipping includes training sample labels being flipped around to lower the judgment of the model. Classification errors can result from label flipping. Label flipping is an example of dirty-label attacks. In contrast, clean-label attacks are created when an adversary makes poisoned training data indistinguishable from nonpoisoned training data. Clean-label attacks are difficult but extremely powerful in terms of detection difficulty. Shafahi et al. [52] stated how in clean-label attacks, the attacker's injected training examples are cleanly labeled by a certified authority, as opposed to maliciously labeled by the attacker herself [52].
6) *Sign-Flipping Attack:* Sign-flipping attacks occur when malicious clients train their local model similar to honest participants, once a normal gradient is obtained, the normal gradient is multiplied by a negative constant when uploading [53], [54]. This attack aims to degrade FL model convergence by uploading misleading gradients with data similarity to honest updates.
7) *Additive Noise Attack:* Additive noise injects noise into the machine learning gradient found within FL local training. Additive noise can increase FL processing times by reducing the quality of local FL updates. Different types of additive noise include Gaussian noise, white noise, impulse noise, and mixed noise which is a combination of different noise.
8) *Backdoor Attack:* A backdoor attack tricks the model into associating a backdoor pattern with a specific target label so that, whenever this pattern appears, the model predicts the target label; otherwise, it behaves normally [55]. Backdoor attacks are also referred to as Trojan attacks. Adversaries inject clean or dirty backdoor updates into data samples in the training phase. When the global model performs aggregation, the model's performance on the specific input is reduced depending on the number of undetected backdoors. When the backdoor is triggered, incorrect predictions occur on the targeted task while the main task performance appears ordinary. Backdoor attacks compromise a subset of samples exchanged with adversaries during local training.

3) Evasion Attacks: An adversary may attempt to evade a deployed system at test time by carefully manipulating attack samples [56]. Evasion attacks can bypass detection during training time with carefully manipulated attack samples. The input looks unaltered to humans but deceiving to the machine learning model. Evasion attacks contaminate training data during aggregation with undetectable misleading data.

4) Byzantine Attacks: In a Byzantine attack, a malicious device gets processed with honest devices. Byzantine attacks aim to harm consensus and decrease model performance. In FL, a malicious attacker may control multiple clients, known as Byzantine users [57]. Byzantine users can upload fake data due to unreliable communication channels, corrupted hardware, or malicious attacks. This leads to the global model being manipulated by attackers and cannot be converged [57]. Larger amounts of Byzantine users increase the effectiveness of Byzantine attacks. In the worst case, a 51% attack could occur when the majority of participants are dishonest.

1) *Sybil Attacks:* Sybil attacks include an adversary controlling multiple pseudonymous identities to increase adversarial influence in a peer-to-peer network. Including multiple fake identities in the participation process can distort model consensus. Fung et al. [58] demonstrated

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TABLE II
SUMMARY OF FL ATTACKS AND RESOURCE COSTS. [REF. = REFERENCE, INTEROP. = INTEROPERABILITY]

Attack type	Ref.	Interop.	Privacy risks	Resource Consumption		Prevention	Impact
				Adversary	Server		
Inference	[44]	High	High	High	Medium	Medium	High
Poisoning	[49]	High	Medium	High	High	Medium	High
Evasion	[56]	High	Low	High	Low	Low	Medium
Byzantine	[61]	Medium	High	High	High	Medium	High
Free-riding	[59]	Low	Low	Low	Medium	High	Medium

that when typical FL convergence heuristics are used, Sybil clones can inflate the training time for as long as they wish, consuming shared resources on both the server and clients [58].

5) Free-Riding Attacks: Within FL environments, the global model is distributed to all participants regardless of their contributions. "Free-riders" may emerge that do not contribute their fair share during the training process. Free-riders create free-rider attacks that lower model performance due to fake model updates being exchanged. A dishonest participant may become a free rider to withhold private information, reduce resource costs, or receive participation rewards while lacking participation requirements. Fraboni et al. [59] mentioned two types of free-riders: 1) plain free-riders, which do not update the local parameters during the iterative federated optimization and 2) disguised free-riders, which employ sophisticated disguising techniques relying on stochastic perturbations of the parameters [59].

1) *Passive Attacks:* Passive attacks are similar to inference attacks in capturing knowledge of the FL network. Passive attacks avoid active modifications to lower detection while acquiring knowledge. Eavesdropping attacks are an example of passive attacks that can go undetected. Passive attacks passively look for vulnerabilities in the network that can be exploited without impacting FL performance.
2) *Active Attacks:* Active attacks commonly follow passive attacks once the network structure is understood. Active attacks directly attack a participant's device, local training, or the online FL model. Nasr et al. [60] experimented with an active attack that performs gradient ascent on a set of target data points before uploading and updating the global parameters. The results magnified the presence of data points in others' training sets, as well as increased the central attacker's inference accuracy.
3) *Timing Attacks:* Timing attacks look at computer hardware and network communications to determine how different inputs impact system response times. Timing attacks look for implementation vulnerabilities in hardware that can reveal hidden information about the network. Timing attacks are similar to side-channel attacks, where internal flaws of a system are attacked to capture information deep within a system.

E. Threats During FL Phases

A range of various attacks can compromise FL security. These attacks can occur during four distinct FL phases: the training phase, parameters exchange phase, parameters aggregation phase, and prediction phase. Each phase has a different attack surface, most susceptible to model poisoning and inference attacks. While FL performs continuous learning, each phase is repeated in multiple iterations. As a result, undetected attacks strengthen as resources are drained from the environment. Four distinct FL phases and potential threats.

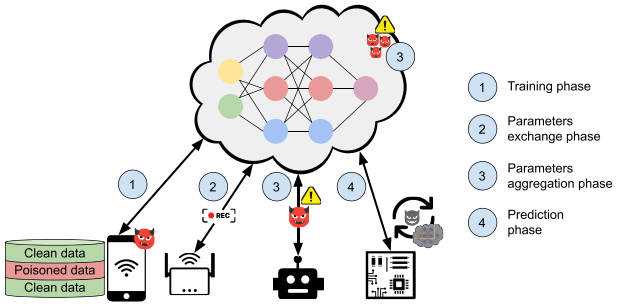


Fig. 4. Potential threat observations within the four phases of FL.

1) *Training Phase:* During the training phase, training data is sent to participants for learning. Local training data is not validated during this phase, allowing adversarial attacks. Early attacks in the training phase occur when the participant's reputation is unavailable. Malicious participants can pollute the training phase by conducting poisoning attacks. Adversaries may target the model's performance to cause the wrong model calculations or lower the target overall training times through misleading data. Increasing amounts of adversarial data negatively impact model effectiveness.
2) *Parameter Exchange Phase:* The parameter exchange phase consists of: a) participants downloading the global model parameters or b) participants uploading the local model parameters. External adversaries may attempt to capture sensitive information during parameter exchange through inference attacks. Inference attacks during this phase can steal parameters (model extraction), combine data to reconstruct the data set, and estimate the distribution of continuously exchanged data. In addition, eavesdropping attacks can occur when communications are intercepted and potentially compromised online.
3) *Parameter Aggregation Phase:* During the parameters aggregation phase, the central server performs a weighted average of participant parameters. Misleading parameters during this phase affect system-wide performance. If the central model architecture is well understood, adversaries can target the central server directly. The central server coordinates the aggregation of online learning and may learn to become malicious due to negative influences such as poisoning attacks. Fig. 4 showcases the danger of adversarial updates being successfully processed into the parameter aggregation phase, thus lowering central server performance. Preserving central server effectiveness is important for the health of the FL architecture. Honest-but-curious central servers attempt to learn all possible information, overstepping boundaries through conducting inference or model poisoning attacks. Inefficient central servers incorrectly redistribute information to participants.

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4) *Prediction Phase:* The prediction phase occurs after the global model has been deployed to devices. Adversaries may continually obtain the deployed global model to perform a combination of evasion and inference attacks. Evasion attacks deceive end devices, causing predictions to be inaccurate. Inference attacks on deployed predictions attempt to extract private information, such as participant data, model parameter data, and training data.

F. Frog-Boiling Attacks in Online Environments

Frog-boiling attacks reveal the limitations of anomaly detection in online environments. Chan-Tin et al. [62] proposed the frog-boiling attack, where an adversary disrupts the network while consistently operating within the threshold of rejection [62]. Frog-boiling refers to the phenomenon that a frog placed in hot water will instantly jump out. In contrast, a frog placed in slightly warm water that gradually heats up will remain in the water and eventually boil to death. Adversaries in online environments may continually attempt smaller attacks to learn the system's state and eventually disrupt the network. Chan-Tin et al. [62] mentioned three variants to the frog-boiling attack: 1) the basic-targeted attack; 2) the network-partition attack; and 3) the closest-node attack [62]. Frog-boiling attacks can avoid outlier detection and increase network-wide latency.

1) Insider and Outsider Attacks: FL environments can be threatened by insider and outsider attacks. Insider attacks can be launched by either the FL server or the participants in the FL system [42]. Insider attacks affect global model aggregation performance and can infer real values from noisy data. On the inside, a dishonest participant may attempt to decode FL updates through malicious tactics. Honest participants on the inside may be impersonated or bribed by external adversaries. In comparison, outsider attacks do not have direct access to internal communications of the network. From the outside, adversaries may eavesdrop on communications between the client and the server. Transferring data over wireless networks can be intercepted, leading to compromised FL updates. Outsiders may control a participant's IoT device through cyberattacks, such as malware, ransomware, and spyware. Nwosu et al. [63] stated cyberattacks mean those attacks launched on online users who are not security conscious either by using a computer network (e.g., spamming, phishing) or as a tool to engage in criminal activities (e.g., cyberstalking or inside man) [63].

2) Device Latency: IoT devices are heterogeneous and contain different resource constraints from traditional computing devices, such as laptops and desktop computers, to cellular devices and online servers. Each device may have different amounts of noise depending on the environment. An FL central server waiting for participants to complete the training phase can be delayed by slower devices. Devices may have unreliable memory bandwidth or unsatisfactory hardware. Delays at the device level can cause delays throughout a synchronous FL architecture. According to Issa et al. [32], the speed of rounds in synchronous FL is restricted to the speed of the slowest device. This causes a "straggler effect," which can cause inefficient processing [32].

TABLE III
EXPANDED TABLE OF FL ATTACKS AND RESOURCE COSTS
[INTEROP. = INTEROPERABILITY]

Attack type	Interop.	Privacy risks	Resource Consumption		Prevention	Impact
			Adversary	Server		
Inference	High	High	High	Medium	Medium	High
Membership	High	High	High	Low	Medium	High
Data properties	High	High	High	Low	Medium	High
Data samples and labels	High	High	High	Low	Medium	High
Model inversion	High	High	High	Low	Low	Medium
Model extraction	Medium	Low	High	Medium	Medium	High
Poisoning	High	Medium	High	High	Medium	High
Model poisoning	High	Low	High	High	Medium	High
Gradient manipulation	High	Low	High	Medium	Low	Medium
Training rule manipulation	High	Low	High	Medium	Low	Medium
Data poisoning	High	Low	High	High	Medium	High
Label manipulation	High	Medium	High	Medium	Medium	Medium
Backdoor updates	Medium	Medium	High	Medium	Medium	Medium
Sign-flipping	High	Low	Medium	High	Medium	Medium
Additive noise	High	Low	Low	Medium	High	Medium
Evasion	High	Low	High	Low	Low	Medium
Byzantine	Medium	High	High	High	Medium	High
Sybil	High	High	High	Medium	Medium	High
Free-riding	Low	Low	Low	Medium	High	Medium
Passive	Low	Low	Low	Low	Low	Low
Active	Low	High	Medium	Medium	Medium	Medium
Timing	Low	Medium	Medium	Medium	Low	Medium

3) Consensus and Dishonesty: Consensus performance is affected by the number of dishonest participants. Aggarwal and Kumar [64] stated that the blockchain network security level is directly proportional to the amount of hash computing power that supports the blockchain. As the miners increase in the mining process, it becomes more difficult for an attacker to attack the blockchain [64]. The amount of honest and dishonest participants directly influence consensus mechanisms in FL and blockchain. Adversaries may employ Sybil attacks, where multiple fake identities are created to throw off consensus mechanisms. The weight of dishonest participants can cause consensus to malfunction. In the worst case, a 51% attack can control network consensus.

4) Additional Noise Against Adversaries: Additional noise creates a tradeoff between levels of security and central server performance. Blanco-Justicia et al. [27] proposed that FL intrinsically protects the data stored on each device by sharing model updates, e.g., gradient information, instead of the original data. However, model updates, which are based on original data, can reveal sensitive information [27]. Adding additional noise can increase the uncertainty of adversaries and reduce adversarial attack effectiveness. Jia et al. [65] proposed to add a carefully crafted noise vector to a confidence score vector to turn it into an adversarial example that misleads the attacker's classifier [65].

Table III contains details on how columns in the expanded table of FL attacks and costs were evaluated and scored. Interoperability was evaluated as the compatibility of attacks in an FL environment. In most cases, adversaries use high amounts of resources to conduct attacks. Some attacks explicitly drain the resources of the central server, while other attacks avoid server resource consumption to evade detection. The prevention column compares detection levels and proposed defenses for specified attacks. The impact column considers the importance of privacy preservation and architecture robustness. Well-planned combinations of attacks increase attack effectiveness.

G. Vulnerabilities of FL and BCFL

Targeted attacks on FL can be reduced by combining with decentralized properties found in blockchain architectures. FL

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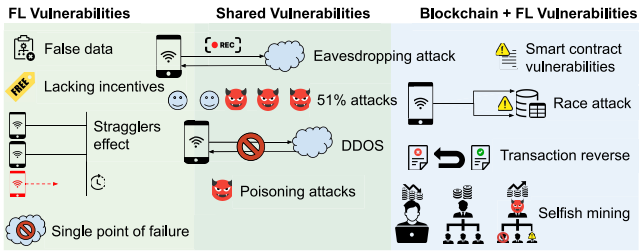


Fig. 5. Various vulnerabilities found in FL and BCFL.

has a large attack surface due to the single point of failure component. An adversary could directly attack the central server to cause instability throughout the FL network. System-wide latency occurs when communications between devices and the central server are unreliable. Blockchain FL offers devices with distributed reliability for computing performance. When a device signals availability, the application layer of the blockchain automates a smart contract for a conditional agreement. Smart contracts on blockchain are immutable, so once created, the smart contract cannot be modified. Imperfect smart contracts include unwanted loopholes or vagueness.

Distributed machine learning architectures, such as BCFL, are vulnerable to outsider attacks while communicating over Internet connections. Communication attacks, such as eavesdropping attacks, man-in-the-middle attacks, and spoofing attacks are potential threats to the exchange of private information. A denial of service on network communications can be highly disruptive. Fig. 5 showcases a range of vulnerabilities found in FL and BCFL. Flooding a network with malicious data is resource-draining for distributed computing architectures.

Adversaries can commit poisoning attacks that reduce performance by sending deceptive inputs to mislead calculations. Dishonest participants redirect optimal resource allocations by providing inaccurate data. Byzantine users who upload fake data can purposely reduce global model performance. A 51% attack occurs when the number of dishonest participants outweighs the number of honest participants; thus, most dishonest participants can gain control of an unbalanced network.

Yang et al. [66] mentioned the cost of creating a 51% attack is surprisingly low if hash power is abundantly available [66]. In FL, a 51% attack can mislead FL training when the majority of participant data is untrustworthy. In blockchain, a 51% attack can control the consensus mechanisms that intervene with transaction authentication. Blockchain transaction authentication can be delayed when conflicting transactions are completed at the same time. A race attack creates a potential fork in the blockchain within the short window of time required to authenticate transactions. Selfish mining pools may purposely generate or withhold blocks simultaneously to gain an advantage over other miners. Transaction reverse is a dishonest mining tactic where the processing of potentially generated blocks is reversed to cause blockchain delays.

H. Integration Motivation of FL and Blockchain

Blockchain has inherent interoperability, allowing it to communicate with other blockchains efficiently. Blockchain interoperability increases robustness when useful information is transferred between blockchains [67]. For example, if a blockchain layer undergoes a denial-of-service attack, a similar blockchain could offer alternative services to supplement security. Similarly, participant data such as reputation can be communicated between blockchains to increase scalability. Blockchain can solve the problem of trust establishment among distributed systems through distributed node verification, and consensus mechanisms [16], [68].

Optimal device selection reduces the dangers of adversaries by removing unreasonable devices from the training process. Soltani et al. [69] mentioned how participation selection in FL has a significant impact on the final model accuracy and training time of FL. Multiple device selection approaches mentioned in [69] can reduce resource costs associated with unfavorable behavior. Ideally, the behavioral patterns between an honest and dishonest participant are largely different. Behavioral auditing of malicious patterns from questionable participants can increase security. Barreno et al. [70] proposed the reject on negative impact (RONI) defense to determine whether a candidate training instance is malicious. Barreno et al. [70] stated if adding the candidate instance to a training set causes the resulting classifier to produce substantially more classification errors, reject the instance as detrimental in its effect.

III. Learning on Resource-Constrained Devices

Latency on IoT devices can increase based on the distance between data centers. Solutions to device latency include moving cloud resources closer to devices, which is seen in mobile edge computing (MEC). Nguyen et al. [36] determined that MEC servers are now becoming a weak point due to data privacy concerns and high data communication overheads. Mobile devices may be subject to attack if not properly secured and may require additional security when deploying important communications in MEC. Li et al. [16] mentioned how blockchain can solve the security problem of edge computing.

IoT devices may have limited resources, and quantifying the expected resource usage can help reduce excess resource expenses [71]. If a device is showing signs of being a straggler, device dropout can be applied to reduce communication costs. However, if we have a large number of stragglers, then simply dropping them can have a negative impact on the overall learning process. Imteaj and Amini [72] developed FedAR algorithm that can select only the effective and trustworthy agents for learning process. Besides, FedPARL framework is specifically designed for resource-constrained FL devices that can prune large models and allow feasible local task allocation for the edge devices [73]. To deal with the heterogeneous resources of the agents, Ahmed et al. [74] proposed an FL model that can generate multiple global models considering the resource status of the agents and accelerate the learning process. Moving forward, increasing device participation can cause diminishing returns if devices perform poorly during training. Adaptive FL can help balance resource costs. Wang et al. [40] proposed an adaptive FL control algorithm based on gradient descent for reducing computational resource budgets. The proposed control algorithm determined the best

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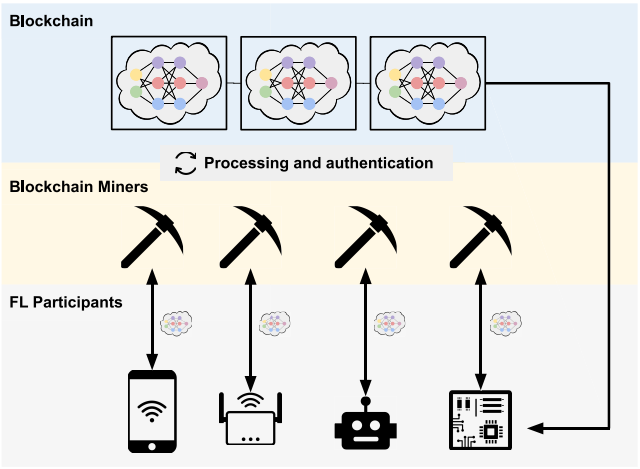


Fig. 6. Condensed workflow with integrated blockchain-based FL.

tradeoff between local update and global parameter aggregation to minimize the loss function under a given resource budget [40].

A. Verification of Local Model Updates

Verifying local model updates helps prevent resource-consuming[1] data from being aggregated with the global model. Wei et al. [5] found that blockchain-assisted decentralized FL frameworks can prevent malicious clients from poisoning the learning process and thus provide a self-motivated and reliable learning environment for clients [5]. Blockchain consensus mechanisms boost network agreement, thus deterring unreasonable data from being aggregated. Yang et al. [75] mention four different blockchain consensus protocols for confirming the correctness of a global model and executing global model aggregation. Blockchain consensus protocols include Proof of Work (PoW), Proof of Stake (PoS), Raft, and practical Byzantine fault tolerance (PBFT) [75].

1) PoW isolates harmful blocks from being added to the network by requiring the completion of a cryptographic puzzle by miners before verification. Lee and Kim [76] proposed that PoW is the most famous algorithm for Bitcoin. Although it works effectively for protection from Sybil attacks and data manipulation, it is expensive due to the required hash power and long block interval [76].
2) PoS randomly selects data validators based on a total stake or overall impact on the network. Individuals with high amounts of holdings are given the opportunity to validate and approve new blocks being added. Validators in PoS create validator nodes to assist in network consensus. The validator's assets are locked while staking to help support network consensus. Validators receive incentives for vouching for the legitimacy of transactions.
3) Raft supports consensus within decentralized blockchain environments. A raft can improve the understandability of the verification by breaking consensus problems down using decomposition. According to Ongaro and Ousterhout [77], Raft separates the key elements of consensus, such as leader election, log replication, and safety, and it enforces a stronger degree of coherency to reduce the number of states that must be considered [77].
4) PBFT ensures a decentralized network continues operating even if a portion of nodes fail or act maliciously. Zhang et al. [78] mentioned that PBFT is an optional consensus protocol for consortium blockchains scenarios, where strong consistency is required [78]. Consortium BCFL can prevent the straggler effect in FL, where the speed of the slowest device causes network-wide delays.

[1]Resource-consuming can be considered negative, harmful, or malicious data.

B. Global Model Aggregation

Global model aggregation in FL occurs iteratively online. Blockchains can provide decentralized applications, such as smart contracts, to achieve global model aggregation without a central server. Fig. 6 displays an example of a FL architecture combined with blockchain in a decentralized manner. The distributed training of the global model can encourage dishonest participants to collude with others and launch a coordinated attack. Bouacida and Mohapatra [79] suggested that past clients can coordinate with current or future participants to participate in attacks against current or future updates to the global model [79]. Global model updates may not change much depending on context. While global model parameters are not changing much, parameters can be frozen to reduce communication costs. Chen et al. [80] found that it is unnecessary to always synchronize the full FL model in the entire training process because many parameters gradually stabilize prior to the ultimate model convergence [80]. Devices with low bandwidth benefit from frozen parameters to reduce device resource costs. Yang et al. [75] proposed a decentralized blockchain-based FL architecture that can resist failures or attacks of servers and devices by building trustworthy global model aggregation with secure model aggregation based on the blockchain consensus protocol among multiple servers.

C. Incentive Mechanism

Blockchain can offer an incentive mechanism in an FL setting to improve participation of clients during the training process. Qu et al. [81] mentioned that there could be heterogeneous devices with different computational and data resources in an FL system. Therefore, participants with better resources must have extra benefits compared to participants with little contribution [81]. Honest contribution is advantageous to all participants and should be rewarded appropriately. Besides, Zhu et al. [82] mentioned how the blockchain's incentive mechanism can track the contribution of each data provider toward the globally optimized model so that participants can be treated fairly, thereby attracting more data sharers. Blockchain-based FL architectures, such as BlockFL [34], enable exchanging devices' local model updates while verifying and providing their corresponding rewards [34]. While Ma et al. [83] found that BCFL enables all clients to verify the learning results that are recorded on the blockchain, whereby distributed clients

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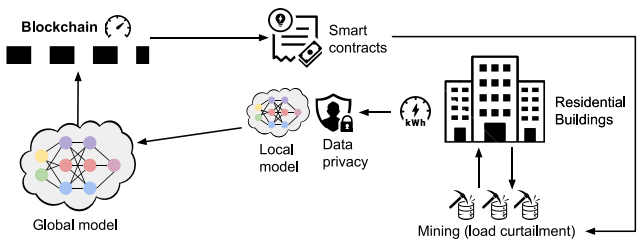


Fig. 7. Example BCFL diagram with an example incentive mechanism for residential buildings.

can be rewarded incentives to participate and untrusted learning models can be detected. Fig. 7 illustrates how a BCFL architecture applied to residential buildings can benefit from blockchain incentive mechanisms.

D. Privacy and Security Protection

Adding noise to data can reduce the overall efficiency. Too much noise negatively affects the performance, while not enough noise may not provide sufficient security guarantees. Levels of noise carry a tradeoff that should be considered. FL can ensure data privacy by only exchanging model parameters with devices. Local device data is never communicated online, boosting the security of data. Multiple approaches examine noise tradeoffs. Truex et al. [39] proposed an FL approach that utilizes both DP and secure multiparty computation (SMC). Furthermore, Truex et al. [39] combined DP with SMC to enable the growth of noise injection to be reduced as the number of parties increases without sacrificing privacy.

Security measures include anticipating anomalous behavior. Typical technologies for defending FL security include malicious participant detection and malicious impact mitigation [51]. Irregular behavior can be indicative of an adversary or potential machine learning attack. Ul Hassan et al. [84] mentioned six anomaly detection models for effectively detecting anomalous behavior: generative architectures, classification-based models, clustering-based models, nearest neighbor models, statistical & analytical models, and reinforcement learning-based models [84]. In addition, participant contribution can be measured to classify behavior between honest versus dishonest participation.

IV. Applications of Blockchain-Integrated Federated Learning

Data-driven technologies can significantly benefit from integrating BCFL. In recent years, there has been an upward trend in applications mentioning blockchain, FL and BCFL. Fig. 8. showcases public publication data from [85]. FL was introduced in 2016, although similar trending topics, such as "Federated Governance" or "Federated Optimization" were captured within the data set. Three benefits of BCFL include inherent privacy preservation, elasticity, and low resource costs. Potential applications, such as smart healthcare, intelligent transportation systems, smart infrastructures, finance, and IoT applications can benefit from including BCFL. Table IV. contains common resource constraints of potential applications.

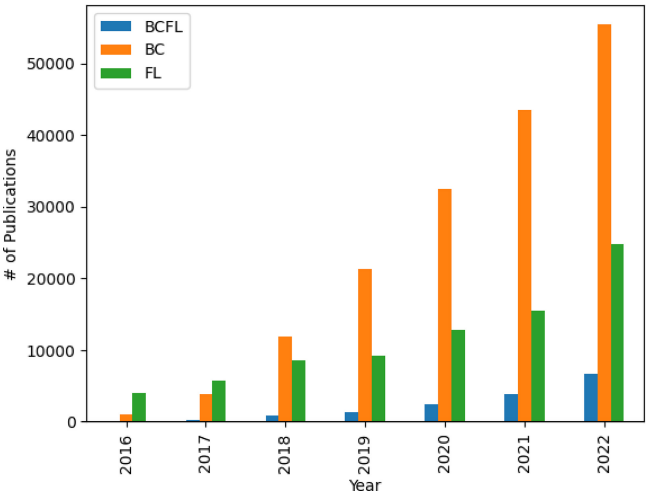


Fig. 8. Number of publications published each year between [2016–2022] that mention "Blockchain," "FL," and "Blockchain FL" [85].

TABLE IV
Resource Constraints of Applications

Resource constraints of applications	BCFL solution
Limited computing power	Increased computing power provided by miners to reduce computing workload.
Network connectivity	Decentralized database with decentralized opportunities.
Privacy preservation	FL privacy preserving mechanisms, such as differential privacy.

A. Smart Healthcare

Smart healthcare environments can benefit from BCFL. FL ensures that machine learning parameters are exchanged compared to personally identifiable information. Nguyen et al. [38] stated that in an FL-based smart healthcare system, only the local updates, such as model gradients, are required by the central server for the AI training, while the local health data are kept at local medical sites and devices. Hai et al. [86] introduced BVFLEMR, a BCFL technology for a cloud-based medical records recommendation system. BCFL can further secure healthcare management. Aich et al. [87] found blockchain FL technologies advantageous for protecting personal healthcare data. FedHealth [10], a transfer FL framework for wearable healthcare, also applies FL for preserving data privacy of personalized healthcare technology.

B. Intelligent Transportation Systems

Intelligent transportation systems process enormous amounts of travel data for coordination. Qi et al. [25] proposed a consortium BCFL framework for traffic flow prediction. Qi et al. [25] developed existing traffic flow prediction methods based on centralized machine learning, which require gathering raw data for model training, and involve serious privacy exposure risks. Heavy haul railways face similar limitations. According to Hua et al. [88], traditional machine learning-based intelligent control methods suffer from insufficient data. Data from different rail lines or operators cannot be shared directly due to the lack of effective incentives and trust [88]. While blockchain may

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provide vehicle integrity and trustworthiness and a reliable data verification tool, FL can be used to preserve privacy, and encryption for confidentiality ensures realistic and implementable solutions that can be implemented into the vehicular network [89].

C. Smart Infrastructures

Smart infrastructures collect data through sensors, process the data, and then employ it for decision making. For instance the security mechanisms of blockchain can improve the reliability of smart infrastructures. In the case of service failure, blockchain can redistribute resources to improve resilience. Otoum et al. [90] suggested that blockchain and FL-enabled solutions provide secure energy trading between different critical entities. Blockchain-based FL can reduce security and trustworthy issues, making it adaptable to the renewable energy trade sector [90]. Feng et al. [31] proposed a horizontal blockchain FL framework for 5G-enabled drones. Cheng et al. [28] mentioned a blockchain FL system for blade icing detection. BLADE effectively reduced the leakage of sensitive business data and avoided failure of the single centralized server by using blockchain [28]. Zhao et al. [41] introduced a blockchain FL system to assist home appliance manufacturers in training a machine learning model based on customers' data. Applications such as smart infrastructures benefit from the privacy preservation and security mechanisms of BCFL. Furthermore, blockchain technology and decentralized opportunities can reduce central resource limitations.

D. Finance and Banking

Transactions within finance and banking may go through multiple intermediaries. Imteaj et al. [15] suggested that long processing times can cause trouble for traders, brokers, regulators, and all other involved parties/stakeholders. Blockchain can resolve this issue by establishing authentication, confidentiality, nonrepudiation through asymmetric crytography, and ensuring the clients' security and privacy through decentralization [15]. Blockchain can offer a secure online trading platform. Intermediaries who may have to fill out monotonous paperwork can instead rely on the self-executing smart contracts of blockchain. FL can also safeguard customer privacy within financial institutions. Internal data silos can be leveraged with FL for data privacy.

E. Internet of Things

IoT applications and how they exchange user data online are related. A group of "things" embedded with software, electronics, actuators, and sensors, which are linked through the Web to gather and swap information with one another [91], is termed as IoT [92]. IoT devices can contribute to FL and blockchain architectures as participants, servers, and overall computing resources.

V. Existing Research Challenges and Future Research Directions

A. Potential Solutions of Existing Research Challenges

Prior works demanded high resource consumption for large-scale data-driven environments. High levels of reliability, data privacy, and distributed computing performance are potential solutions when FL is coupled with Blockchain. Privacy-preserving mechanisms such as DP can encourage resourceful participation from different IoT devices. Resourceful FL environments generally have two main challenges.

1) High communication costs can be discouraged based on network reliability. Communication delays from service providers may require a device to drop out at slower speeds. Cui et al. [30] examined compressed communication to reduce communication overheads in BCFL. They developed a Top_k sparsification communication-efficient framework that could reduce the training time by about 95% without compromising model accuracy. Approaches that reduce communication overheads are advantageous for lowering participation requirements. Top_k sparsification compresses communications within distributed machine learning environments by selecting the most significant k elements from gathered scores. Similar communication compression solutions include gradient compression, thresholding, quantization, and parameter averaging.
2) Tradeoffs between security and performance are existing resource challenges for BCFL. On one side, increasing noise against participants boosts security and lowers performance. On the other side, allowing open participation can increase available computing resources, but participants are not fully trusted. Choosing the best devices for participation can reduce the dependence on tradeoffs between security and performance. Private FL and consortium blockchains are present solutions for the possible diminishing returns of open participation. Permissioned approaches enable screening of participating devices to encourage optimal participation selection of trusted devices. Mohammed et al. [93] proposed a budgeted number of candidate clients chosen from the best candidate clients in terms of test accuracy to participate in the training process. Optimal selection strategies can verify participants are resourceful and honest before inviting these participants into training. Blockchain technology can store participation records in its distributed ledger. Communicating reputation or participation records between blockchains can improve scalability. Dishonest participation records are not shared between blockchains. Records of dishonest behavior can warn similar blockchains of a participant's integrity. For example, a dishonest mining pool with evidence of performing mining attacks against blockchain participants can be blacklisted. Blockchain interoperability can allow participants to be chosen thoughtfully based on their assumed trust or available resources.

By fostering participation and boosting overall resource efficacy, BCFL has the potential to improve resource-constrained

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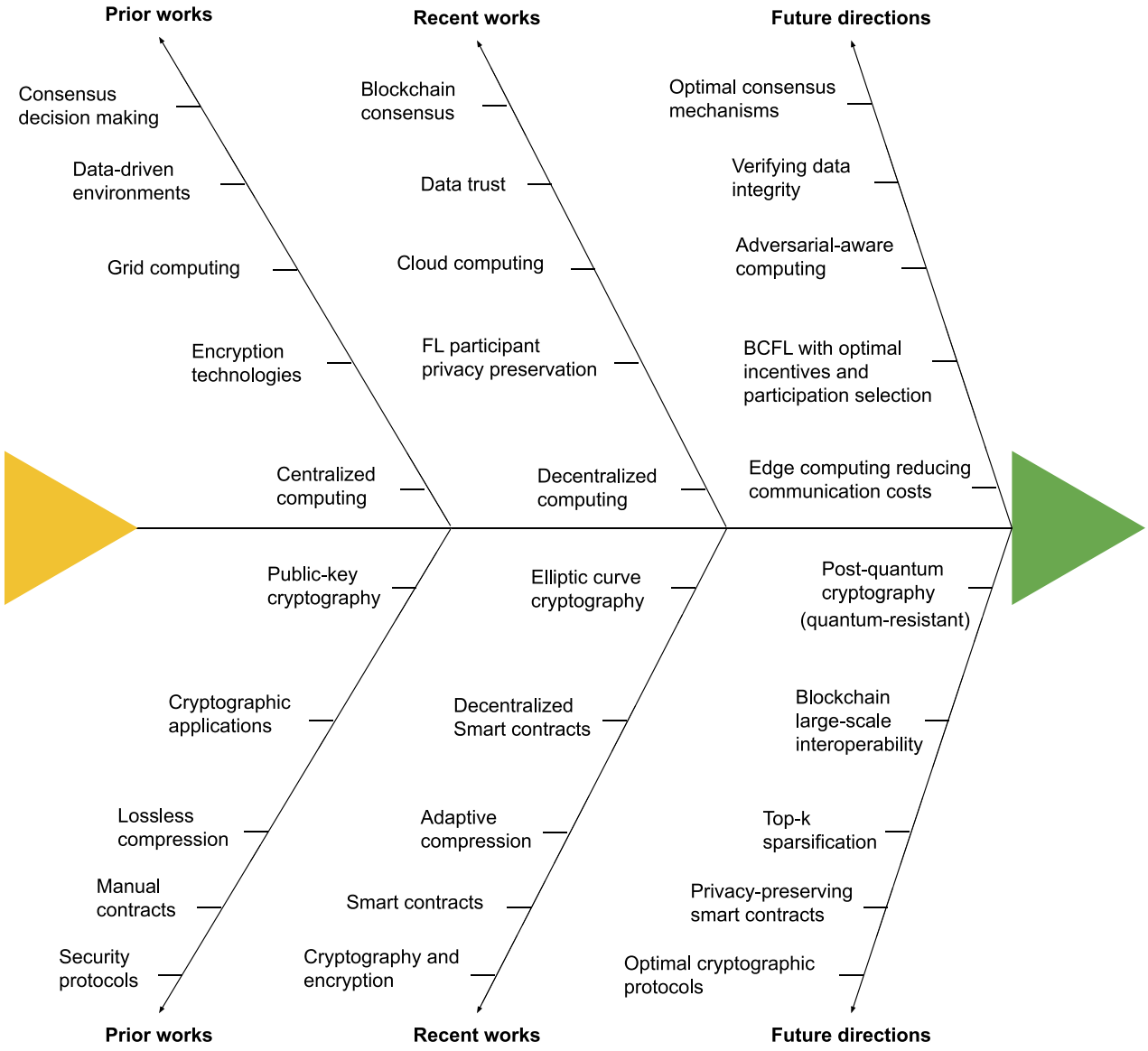


Fig. 9. BCFL flow from prior works toward future directions.

environments. Li et al. [35] demonstrated that blockchain technology not only improves the security and performance of FL but also broadens its applicability to other domains.

B. Future Research Directions

We identify six areas for future research in BCFL: 1) adversarial aware defenses; 2) optimal incentives and participation selection; 3) edge computing; 4) post-quantum cryptography; 5) blockchain interoperability; 6) compressed communications; 7) optimal consensus mechanisms; 8) Data integrity; 9) privacy-preserving smart contracts; and 10) optimized cryptographic protocols. Each of these areas contributes to the alignment of secure and private BCFL environments. Fig. 9 displays a timeline of important BCFL research directions. Future directions such as being quantum resistant can disrupt FL resource management if implemented slowly or incorrectly.

1) Adversarial aware computing includes defense mechanisms within the architecture to protect the network from adversarial attacks such as poisoning attacks. The FL server side could benefit from included defenses such as GANs found in [94]. Zhao et al. [94] proposed a novel poisoning defense generative adversarial network (PDGAN), which reconstructs training data from model updates and audits the accuracy for each participant model using the generated data. Evaluating participant updates further protects the server side, although participants could be compromised. To lessen adversarial attempts on participant devices, participants could undergo authentication such as two-factor authentication (2FA). Xu et al. [95] combined blockchain technology and FL for dynamic identity management along with verifying the legitimacy of the identity of participants.
2) Optimal incentives and participation selection can benefit available resources in a BCFL environment. Potential participants supply as miners or network providers.

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Thoughtful incentives can encourage the cooperation of prosumers. Blockchain mining pools may interfere with other mining pools, based on proposed incentives. Nojoumian et al. [96] proposed a blockchain reputation-based mining paradigm for encouraging honest mining. The trust management schema discouraged dishonest mining techniques, such as selfish mining, stubborn mining, and eclipse attacks. Furthermore, the proposed framework included theory to prevent re-entry attacks, where corrupted players return to the scheme using new identities [96].

3) Edge computing can reduce network costs for far away nodes by moving computing resources closer to devices. IoT devices that may need to dropout of the network due to latency, computing speeds, or storage size can move closer to edge devices. Incorporating potential edge devices in the network can reduce disruptions between participants and the central server. In such a case, resourceful participants or mining pools can act as an edge device. Still, blockchain scalability and overall size can exceed available storage for IoT devices with limited storage size. Imteaj et al. [15] stated that as the number of transactions increases, it requires a larger blockchain size to store those transactions. However, mining a large size of blockchain may require more resources, which would be difficult for IoT devices [15]. Applying dimensional reduction for unreasonably large blockchains can prevent future IoT participants from requiring supercomputers, compared to current GPUs.
4) Post-quantum cryptography algorithms may be required for resilience from quantum attacks. Fernández-Caramès and Fraga-Lamas [97] mentioned how the fast progress of quantum computing has opened the possibility of performing attacks based on Grover's and Shor's algorithms. Such algorithms threaten public-key cryptography, and hash functions [97]. Blockchain may require a cryptographic redesign to combat quantum attacks. Quantum computers bring unique threats to classical data management techniques.
5) Blockchain interoperability can build upon the foundation of cryptographic applications. Blockchain transactions operate on smart contracts, which are self-executing protocols. Blockchain technology can apply cross-chain communication protocols to transfer block information seamlessly. If a blockchain is directly attacked or low on resources, a similar blockchain could transfer network resources using blockchain interoperability. Belchior et al. [98] mentioned how Blockchain of Blockchains are frameworks that provide reusable data, network, consensus, incentive, and contract layers for the creation of application-specific blockchains (customized blockchains) that interoperate between each other.
6) Compressed communications can reduce the communication costs of large networks. Communications require energy consumption, bandwidth, and may overall be congested depending on demand. Reducing data transmissions can lower latency and improve overall scalability. In FL, the communication overhead of each client can be reduced by extracting high-ranking parameters from the local gradient. Lu et al. [99] proposed a Top_k sparsification FL framework that could reduce communication overhead and wall clock training time. Top_k sparsification can be applied to BCFL architectures for compressed communications [30].
7) Innovative consensus mechanisms need to be designed considering the special learning architecture of FL, and further research needs to be conducted to ensure secure and privacy-preserved model aggregation enabling consensus mechanisms [e.g., PoS, Proof of Authority (PoA), Proof of Burn (PoB)]. For instance, PoB is a prospective solution to reduce the high energy consumption found in PoW algorithms. Karantias et al. [100] discuss PoB verification mechanisms with different security guarantees.
8) Exploring the potential of ensuring data integrity and trust in an FL environment tailoring the decentralized nature of blockchain while minimizing communication overhead is of paramount importance. For instance, integrating end-to-end encryption or self-destructing data transmissions can protect communication confidentiality [101]. Additionally, the use of metrics could create a distinct threshold for differentiating between honest, dishonest, and honest-but-curious model changes, strengthening the system's robustness.
9) Privacy-preserving smart contracts need to be developed to ensure the confidentiality of the model updates while allowing verifiable and reliable aggregation examining the cutting-edge cryptographic methods including homomorphic encryption, safe multiparty computing, and zero-knowledge proofs [102].
10) It is crucial to develop the FL framework enabling blockchain technology and extend its applications for resource-constrained users, such as edge computing nodes and IoT devices [103]. In order to achieve this goal, researchers need to come up with optimization methods and create simple cryptographic protocols that can fit the limited processing and storage capabilities of such devices.

VI. Conclusion

In this article, we present legible insights into blockchain-based FL from crucial and time-demanding perspectives. Blockchain provides additional security in a decentralized format that protects FL from various real-world security and privacy issues. We categorize various threat models and point out the leading vulnerabilities that could be observed during various stages of the FL process and in online environments of blockchain-enabled FL settings. To this end, we show a clear direction on how we can leverage secure and private learning on resource-constrained devices, covering the feasible verification of local model updates, secure global model aggregation, designing fair incentive schemes, and upgrading security and protection. Finally, we present the existing blockchain-based FL applications and highlight the potential solutions to the existing research challenges in the relevant

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domains. We anticipate that this survey will be helpful for researchers, practitioners, and scientists, developing robust blockchain-enabled FL systems choosing suitable consensus mechanisms, identifying security and privacy pitfalls, motivating coherent formations, and following the promising future directions presented in this article.

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